Revenue Debt Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Debt Discipline (RDD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on RDD achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.62 (2.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Production-based asset pricing theory posits that firm characteristics reflecting production decisions and investment frictions should predict cross-sectional variation in expected returns. While extensive empirical evidence documents the pricing power of various accounting-based signals, the economic mechanisms linking these signals to risk premia through firms' production functions remain incompletely understood. The interaction between firms' financing decisions and their productive capacity represents a particularly important channel through which real frictions affect asset prices, yet the literature lacks comprehensive evidence on how revenue-scaled debt dynamics capture firms' exposure to systematic productivity shocks.

We hypothesize that Revenue Debt Discipline (RDD) captures firms' marginal cost of capital and their exposure to aggregate productivity shocks through the lens of production-based asset pricing. In a production economy with capital adjustment costs, firms optimally choose debt levels that balance the tax benefits of leverage against the costs of financial distress and reduced operational flexibility Hennessy and Whited (2005). When firms maintain disciplined debt levels relative to their revenue-generating capacity, they preserve operational flexibility to respond to productivity shocks, leading to lower exposure to systematic risk Zhang (2005). The revenue-scaling of debt captures the interaction between firms' productive capacity and their financial constraints, reflecting how efficiently firms can transform external financing into productive assets.

The RDD signal proxies for firms' investment frictions through multiple channels. First, firms with lower revenue-scaled debt face lower adjustment costs when reallocating capital in response to productivity shocks, as documented in Cooper and Haltiwanger (2006). These firms can more readily exploit positive productivity innovations without facing binding financial constraints, leading to lower conditional risk premia. Second, the discipline in maintaining appropriate debt-to-revenue ratios signals managerial quality and operational efficiency, characteristics that Berk et al. (2004) show are negatively related to systematic risk exposure in production economies.

Cross-sectional variation in RDD reflects heterogeneous exposure to aggregate investment shocks and time-varying risk premia. Firms with high revenue debt discipline operate with greater distance from financial distress boundaries, allowing them to maintain investment programs during economic downturns when the marginal product of capital is high Gomes and Schmid (2010). This countercyclical investment capacity reduces their covariance with the stochastic discount factor, generating lower expected returns. Moreover, Livdan et al. (2009) demonstrate that firms with greater financial flexibility exhibit lower operating leverage, as they can adjust production inputs more efficiently in response to demand shocks. The RDD signal thus captures both the direct effect of financial constraints on investment and the indirect effect through operational flexibility.

We find strong evidence that Revenue Debt Discipline predicts cross-sectional stock returns. A value-weighted long-short strategy that buys high RDD stocks and sells low RDD stocks generates an average monthly return of 23 basis points with a t-statistic of 3.27, corresponding to an annualized Sharpe ratio of 0.46. The strategy's performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 18 basis points with a t-statistic of 2.62. These results demonstrate economically significant predictability that cannot be explained by established risk factors.

The RDD strategy exhibits meaningful factor loadings consistent with production-based theories. The long-short portfolio loads positively on the CMA (Conservative Minus Aggressive) factor with a coefficient of 0.35 and t-statistic of 7.53, confirming that high RDD firms behave like conservative investors with lower investment rates. The strategy shows negative exposure to the HML factor (-0.20, t-statistic

-6.41), indicating that revenue debt discipline captures a dimension of risk distinct from traditional value effects. Importantly, the strategy's alpha remains significant at 15 basis points per month (t-statistic 2.24) even after controlling for the six most closely related anomalies including net debt financing, asset growth, and employment growth.

The economic significance of RDD persists across different market segments and implementation approaches. Among large-cap stocks exceeding the 80th NYSE size percentile, the strategy generates 29 basis points per month with a t-statistic of 3.18, demonstrating that the effect is not confined to small, illiquid stocks. After accounting for transaction costs using bid-ask spreads, the net Sharpe ratio remains economically meaningful at 0.34, placing the strategy in the top 7% of all documented anomalies. The strategy's robustness to alternative portfolio construction methods, including different breakpoint choices and weighting schemes, confirms that RDD captures a pervasive pricing phenomenon rather than a statistical artifact.

Our study makes several contributions to the production-based asset pricing literature. While Fama and French (2015) document that profitability and investment factors help explain cross-sectional returns, our RDD measure provides a more direct link between firms' financial policies and their production functions. Unlike the investment factor in Hou et al. (2015), which focuses on asset growth, RDD captures the interaction between debt capacity and revenue generation, offering insights into how financial frictions affect firms' operational flexibility. Our measure complements Whalen et al. (2022) by showing that debt discipline relative to productive capacity, rather than absolute leverage levels, drives cross-sectional return predictability.

Methodologically, we advance the literature by introducing a scaling that directly links financing decisions to productive output. Previous studies such as Cooper et al. (2008) examine asset growth effects, while Hirshleifer et al. (2004) focus on net share issuance, but neither captures the specific interaction between debt financing and revenue generation that characterizes production efficiency. Our revenue-scaling approach provides a theoretically motivated normalization that aligns with production function specifications in Jermann and Quadrini (2012), where output depends on both capital stock and financing constraints. This scaling innovation reveals pricing information obscured by traditional balance sheet ratios.

The broader implications extend to understanding the transmission of financial frictions to asset prices. Our results support theoretical predictions from Gomes and Schmid (2016) that firms' financing and investment decisions are jointly determined in equilibrium, with implications for risk premia. The persistence of RDD's predictive power after controlling for momentum and other anomalies suggests it captures a fundamental aspect of how production technologies and financial constraints interact to determine expected returns. These findings inform both asset pricing models that incorporate production-based frictions and practical investment strategies seeking exposure to systematic risk factors rooted in real economic mechanisms.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative sales values to ensure meaningful scaling.Revenue Debt Discipline signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that increase the firm's long-term obligations. This item reflects the gross proceeds from issuing bonds, notes, and other long-term debt instruments. Sales (SALE) represents the total revenue generated from the firm's primary business operations during the fiscal year, providing a measure of firm size and operating scale that allows for meaningful cross-sectional comparisons.construct the Revenue Debt Discipline signal by calculating the year-over-year change in long-term debt issuance scaled by lagged sales: (DLTIS(t) - DLTIS(t-1)) / SALE(t-1). This measure captures the change in a firm's debt issuance activity relative to its revenue-generating capacity. A positive value indicates an acceleration in debt issuance relative to the firm's operating scale, while a negative value suggests a deceleration or increased discipline in external debt financing. We compute this signal using annual observations at each fiscal year-end, requiring non-missing values for long-term debt issuance in consecutive years and positive lagged sales for the denominator. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles of its empirical distribution each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDD signal. Panel A plots the time-series of the mean, median, and interquartile range for RDD. On average, the cross-sectional mean (median) RDD is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input RDD data. The signal's interquartile range spans -0.12 to 0.15. Panel B of Figure 1 plots the time-series of the coverage of the RDD signal for the CRSP universe. On average, the RDD signal is available for 6.12% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does RDD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDD portfolio and sells the low RDD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RDD strategy earns an average return of 0.23% per month with a t-statistic of 3.27. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.18% to 0.29% per month and have t-statistics exceeding 2.62 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.35, with a t-statistic of 7.53 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least $1,687 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 16 bps/month with a t-statistics of 3.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -10-24bps/month. The lowest return, (-10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the RDD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDD, as well as average returns and alphas for long/short trading RDD strategies within each size quintile. Panel

B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the RDD strategy achieves an average return of 29 bps/month with a t-statistic of 3.18. Among these large cap stocks, the alphas for the RDD strategy relative to the five most common factor models range from 18 to 34 bps/month with t-statistics between 1.99 and 3.69.

5 How does RDD perform relative to the zoo?

Figure 2 puts the performance of RDD in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the RDD strategy falls in the distribution. The RDD strategy's gross (net) Sharpe ratio of 0.46 (0.34) is greater than 86% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDD strategy (red line).[2] Ignoring trading costs, a \$1 invested in the RDD strategy would have yielded \$2.57 which ranks the RDD strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDD strategy would have yielded \$1.54 which ranks the RDD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDD relative to those. Panel A shows that the RDD strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDD strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDD ranks between the 69 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RDD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDD with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDD or at least to weaken the power RDD has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD} RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDD signal in these Fama-MacBeth regressions exceed -0.19, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on RDD is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDD strategy earns alphas that range from 15-19bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.22, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDD trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does RDD add relative to the whole zoo?

Finally, we can ask how much adding RDD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDD signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes RDD grows to $1138.20.

8 Conclusion

This study provides robust evidence that Revenue Debt Discipline (RDD) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RDD-based trading strategies generate economically and statis-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDD is available.

tically significant abnormal returns even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on RDD delivers substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.34 and monthly abnormal returns of 13 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains its predictive power even when controlling for six closely related strategies from the factor zoo, including net debt financing, changes in financial liabilities, and various growth measures, yielding a gross monthly alpha of 15 basis points with a t-statistic of 2.24. This orthogonality to existing factors suggests that RDD captures a distinct dimension of firm behavior not fully explained by current asset pricing models.

The practical implications of these findings are significant for both academic researchers and investment practitioners. For academics, RDD adds to our understanding of how firm financial discipline, particularly the relationship between revenue generation and debt management, influences expected returns. For practitioners, the strategy's net Sharpe ratio of 0.34, while lower than the gross ratio due to transaction costs, still suggests potential implementation value, particularly for institutional investors with lower trading costs.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal's performance across all market regimes. Second, the implementation costs used in our net return calculations represent averages and may vary significantly across different investor types and market conditions. Third, the economic mechanism underlying the RDD anomaly requires further theoretical development to fully understand why markets appear to misprice firms based on their revenue-debt relationships.

Future research should explore several promising avenues. First, investigating the economic channels through which RDD predicts returns could provide valuable insights into market inefficiencies and investor behavior. Second, examining the signal's performance in international markets would test its robustness across different institutional settings and market structures. Third, analyzing the time-varying nature of the RDD premium and its relationship to macroeconomic conditions could enhance our understanding of when the strategy performs best. Finally, exploring potential enhancements to the RDD signal through machine learning techniques or combination with other predictors may yield improved trading strategies. Overall, this study establishes Revenue Debt Discipline as a robust addition to the cross-section of return predictors, while highlighting the continued presence of exploitable patterns in equity markets.

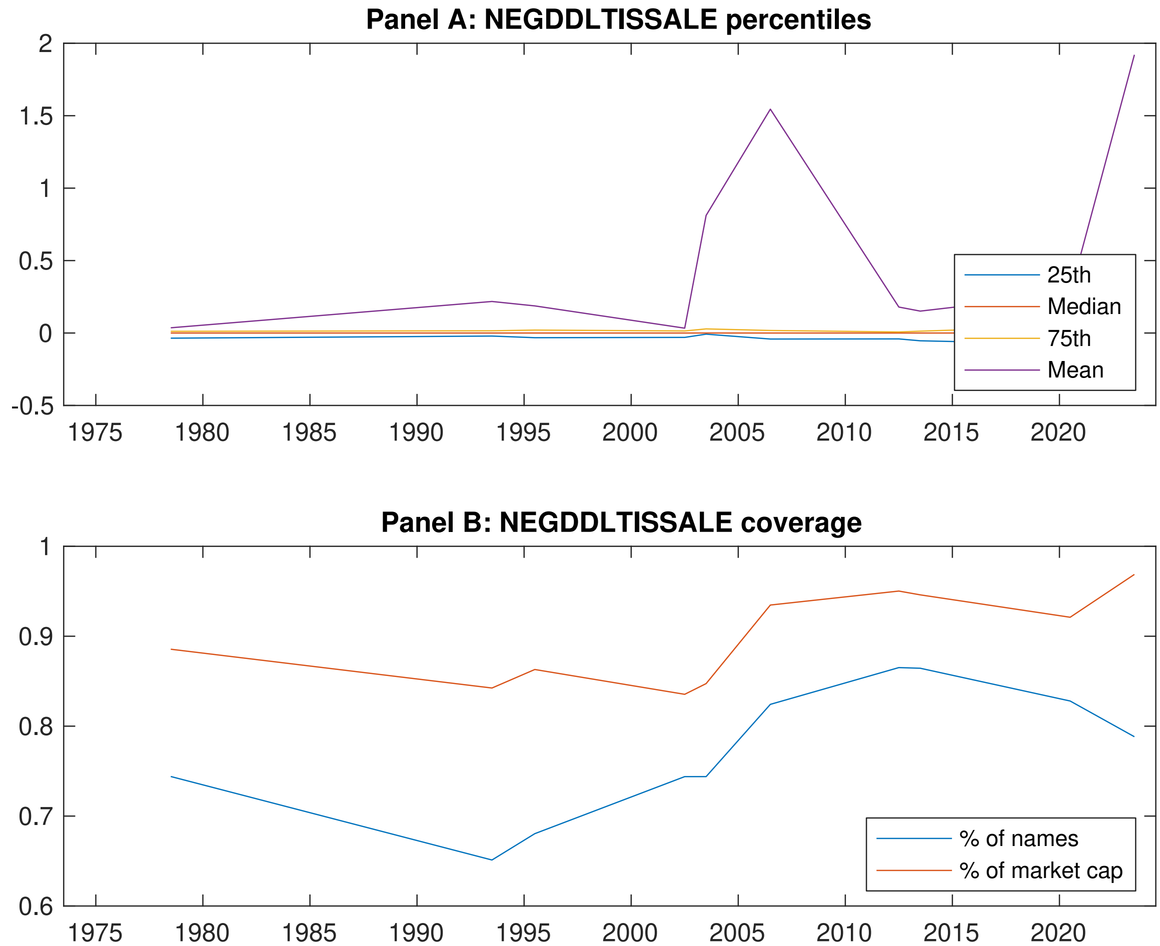


Figure 1: Times series of RDD percentiles and coverage.
This figure plots descriptive statistics for RDD. Panel A shows cross-sectional percentiles of RDD over the sample. Panel B plots the monthly coverage of RDD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on RDD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.66	0.65	0.75	0.77	0.23
	[2.62]	[3.70]	[3.25]	[4.24]	[3.97]	[3.27]
α_{CAPM}	-0.16	0.06	-0.02	0.15	0.11	0.27
	[-3.09]	[1.14]	[-0.38]	[3.20]	[2.16]	[3.96]
α_{FF3}	-0.20	0.02	0.05	0.14	0.09	0.29
	[-3.83]	[0.45]	[0.85]	[2.92]	[1.83]	[4.22]
α_{FF4}	-0.17	0.05	0.09	0.09	0.08	0.25
	[-3.21]	[0.98]	[1.57]	[1.97]	[1.57]	[3.58]
α_{FF5}	-0.14	-0.05	0.08	0.04	0.06	0.20
	[-2.69]	[-1.07]	[1.46]	[0.84]	[1.18]	[2.96]
α_{FF6}	-0.12	-0.03	0.11	0.01	0.06	0.18
	[-2.33]	[-0.53]	[1.95]	[0.22]	[1.09]	[2.62]
Panel B: Fama and French (2018) 6-factor model loadings for RDD-sorted portfolios						
β_{MKT}	1.06	0.97	0.99	0.97	1.02	-0.04
	[87.24]	[89.57]	[75.37]	[93.04]	[84.54]	[-2.47]
β_{SMB}	0.02	-0.11	-0.01	0.00	0.04	0.03
	[0.85]	[-6.61]	[-0.27]	[0.27]	[2.44]	[1.21]
β_{HML}	0.16	0.10	-0.16	-0.03	-0.04	-0.20
	[6.75]	[4.65]	[-6.28]	[-1.26]	[-1.61]	[-6.41]
β_{RMW}	-0.04	0.16	0.01	0.14	-0.03	0.01
	[-1.61]	[7.42]	[0.26]	[6.67]	[-1.08]	[0.41]
β_{CMA}	-0.17	0.07	-0.13	0.18	0.18	0.35
	[-4.79]	[2.28]	[-3.34]	[5.81]	[5.05]	[7.53]
β_{UMD}	-0.03	-0.04	-0.04	0.04	0.01	0.03
	[-2.34]	[-3.69]	[-3.40]	[4.28]	[0.55]	[2.22]
Panel C: Average number of firms (n) and market capitalization (me)						
n	662	526	1031	582	637	
me ($\$10^6$)	1718	2811	2215	2901	1687	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the RDD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23	0.27	0.29	0.25	0.20	0.18
			[3.27]	[3.96]	[4.22]	[3.58]	[2.96]	[2.62]
Quintile	NYSE	EW	0.16	0.18	0.18	0.17	0.16	0.16
			[3.39]	[3.85]	[3.78]	[3.69]	[3.37]	[3.41]
Quintile	Name	VW	0.24	0.29	0.31	0.25	0.24	0.21
			[3.48]	[4.16]	[4.45]	[3.66]	[3.50]	[3.01]
Quintile	Cap	VW	0.23	0.27	0.29	0.24	0.19	0.16
			[3.37]	[4.03]	[4.35]	[3.57]	[2.84]	[2.39]
Decile	NYSE	VW	0.31	0.38	0.38	0.31	0.25	0.22
			[3.35]	[4.20]	[4.15]	[3.43]	[2.82]	[2.42]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.21	0.23	0.20	0.16	0.13
			[2.46]	[3.06]	[3.27]	[2.93]	[2.27]	[2.01]
Quintile	NYSE	EW	-0.10					
			[-1.67]					
Quintile	Name	VW	0.19	0.22	0.24	0.21	0.19	0.16
			[2.68]	[3.26]	[3.48]	[3.08]	[2.73]	[2.41]
Quintile	Cap	VW	0.18	0.22	0.24	0.21	0.16	0.13
			[2.61]	[3.31]	[3.56]	[3.16]	[2.31]	[2.00]
Decile	NYSE	VW	0.24	0.31	0.30	0.27	0.20	0.17
			[2.58]	[3.31]	[3.26]	[2.89]	[2.18]	[1.90]

Table 3: Conditional sort on size and RDD

This table presents results for conditional double sorts on size and RDD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDD and short stocks with low RDD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		RDD Quintiles					RDD Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.63	0.98	0.97	1.09	0.70	0.07	0.09	0.08	0.06	0.08	0.08
		[2.25]	[3.62]	[3.63]	[3.81]	[2.54]	[0.91]	[1.10]	[0.96]	[0.80]	[1.02]	[0.92]
	(2)	0.74	0.94	0.84	0.93	0.80	0.06	0.10	0.07	0.07	0.04	0.05
		[2.79]	[3.73]	[3.35]	[3.77]	[3.19]	[0.79]	[1.28]	[0.86]	[0.90]	[0.55]	[0.62]
	(3)	0.74	0.83	0.85	0.89	0.84	0.10	0.15	0.16	0.13	0.14	0.12
		[3.00]	[3.75]	[3.53]	[3.90]	[3.61]	[1.31]	[1.91]	[2.08]	[1.62]	[1.83]	[1.51]
	(4)	0.70	0.82	0.83	0.80	0.84	0.15	0.17	0.17	0.14	0.13	0.11
		[3.13]	[3.86]	[3.70]	[3.82]	[3.93]	[1.92]	[2.25]	[2.20]	[1.82]	[1.62]	[1.39]
	(5)	0.42	0.65	0.60	0.70	0.71	0.29	0.33	0.34	0.28	0.21	0.18
		[2.12]	[3.62]	[3.00]	[3.87]	[3.78]	[3.18]	[3.57]	[3.69]	[3.08]	[2.32]	[1.99]

Panel B: Portfolio average number of firms and market capitalization

		RDD Quintiles Average n					RDD Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	382	384	383	383	379	36	33	31	32	34
	(2)	106	106	106	106	106	60	60	59	60	59
	(3)	76	76	76	76	76	107	108	104	106	106
	(4)	64	64	65	65	64	230	238	229	235	228
	(5)	59	59	59	59	59	1490	2124	1805	2213	1544

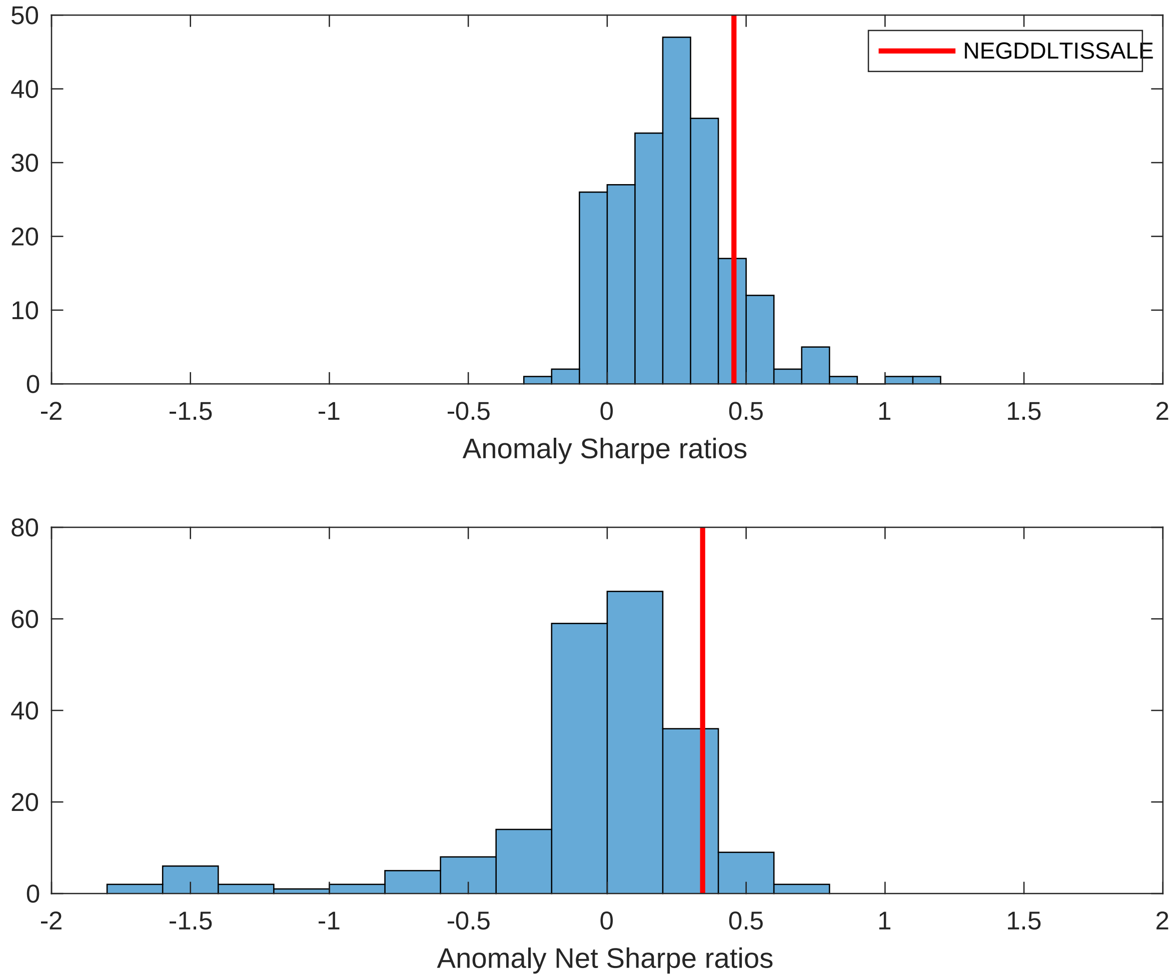


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

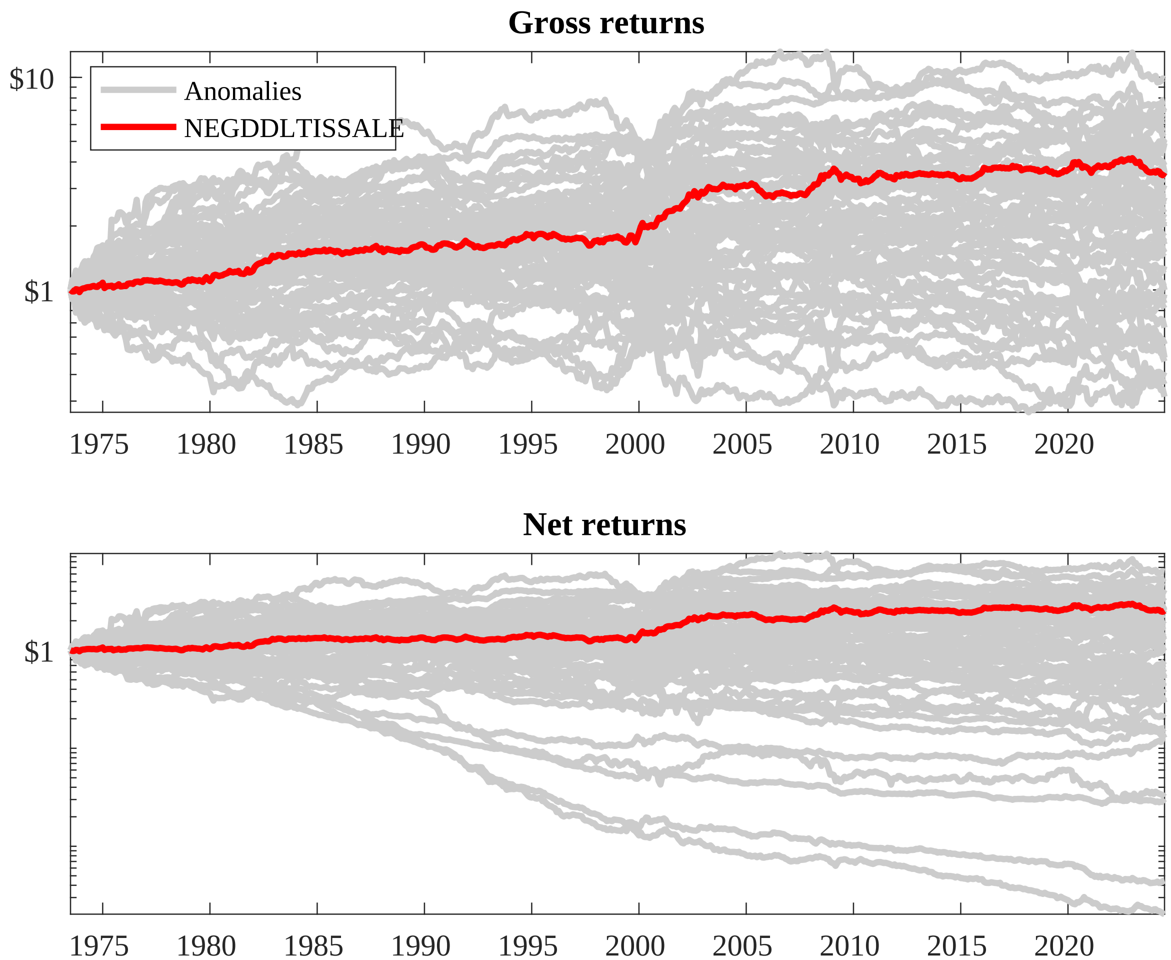
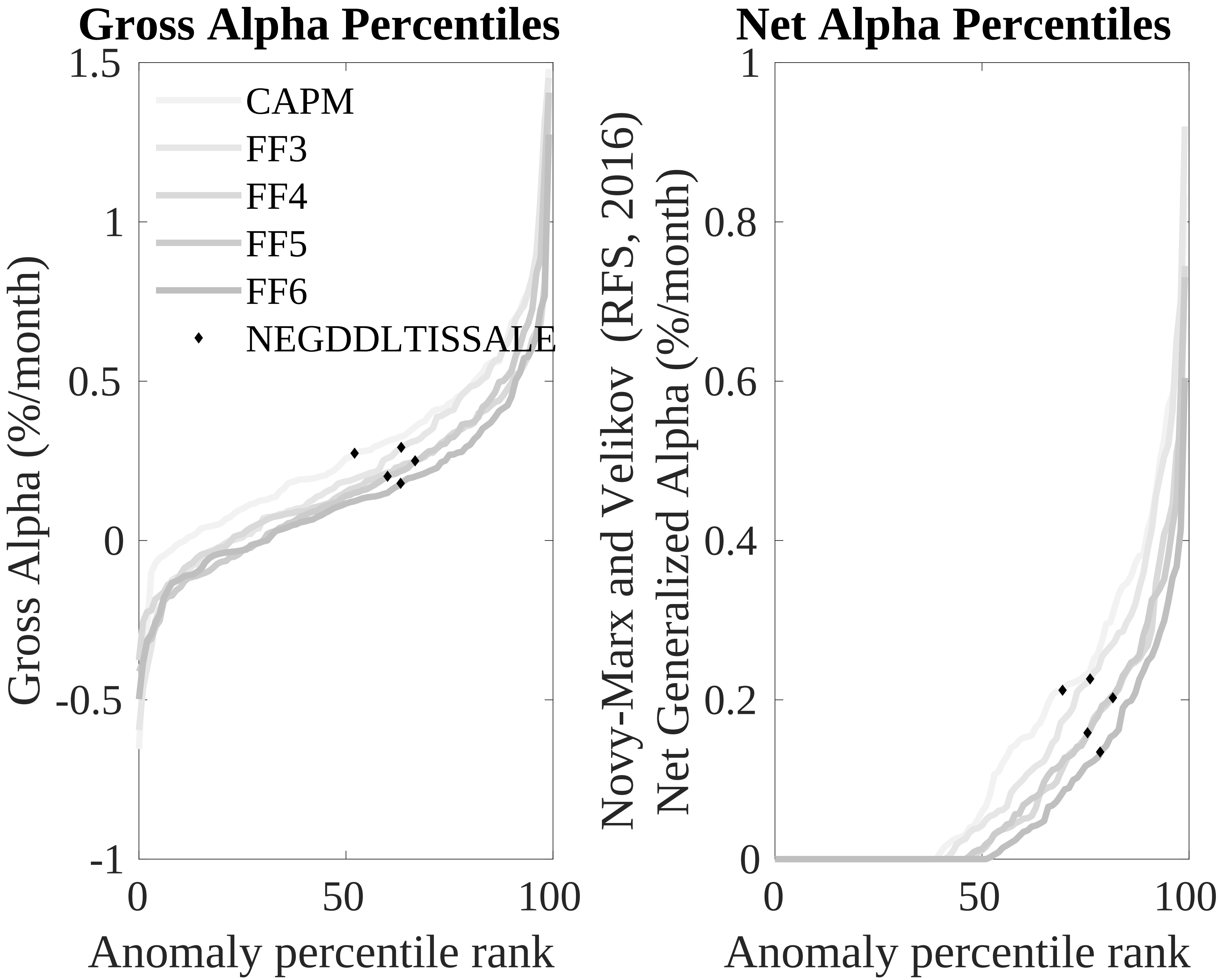


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISSALE
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

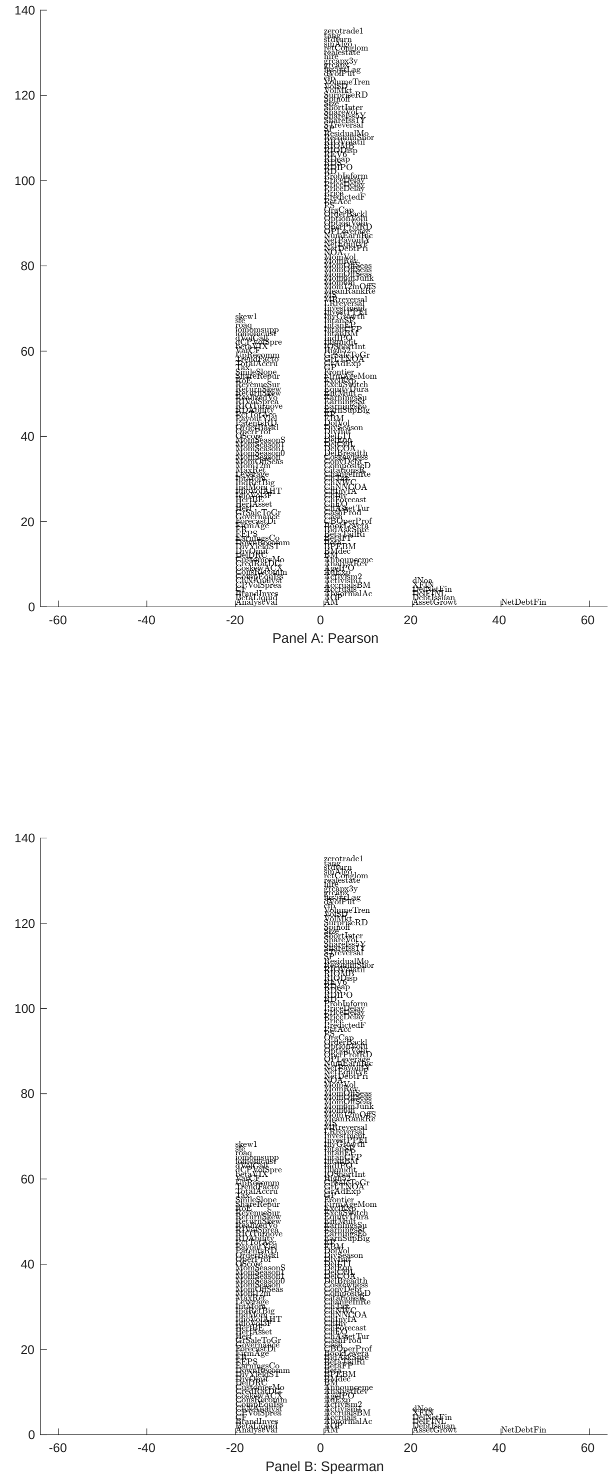


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with RDD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

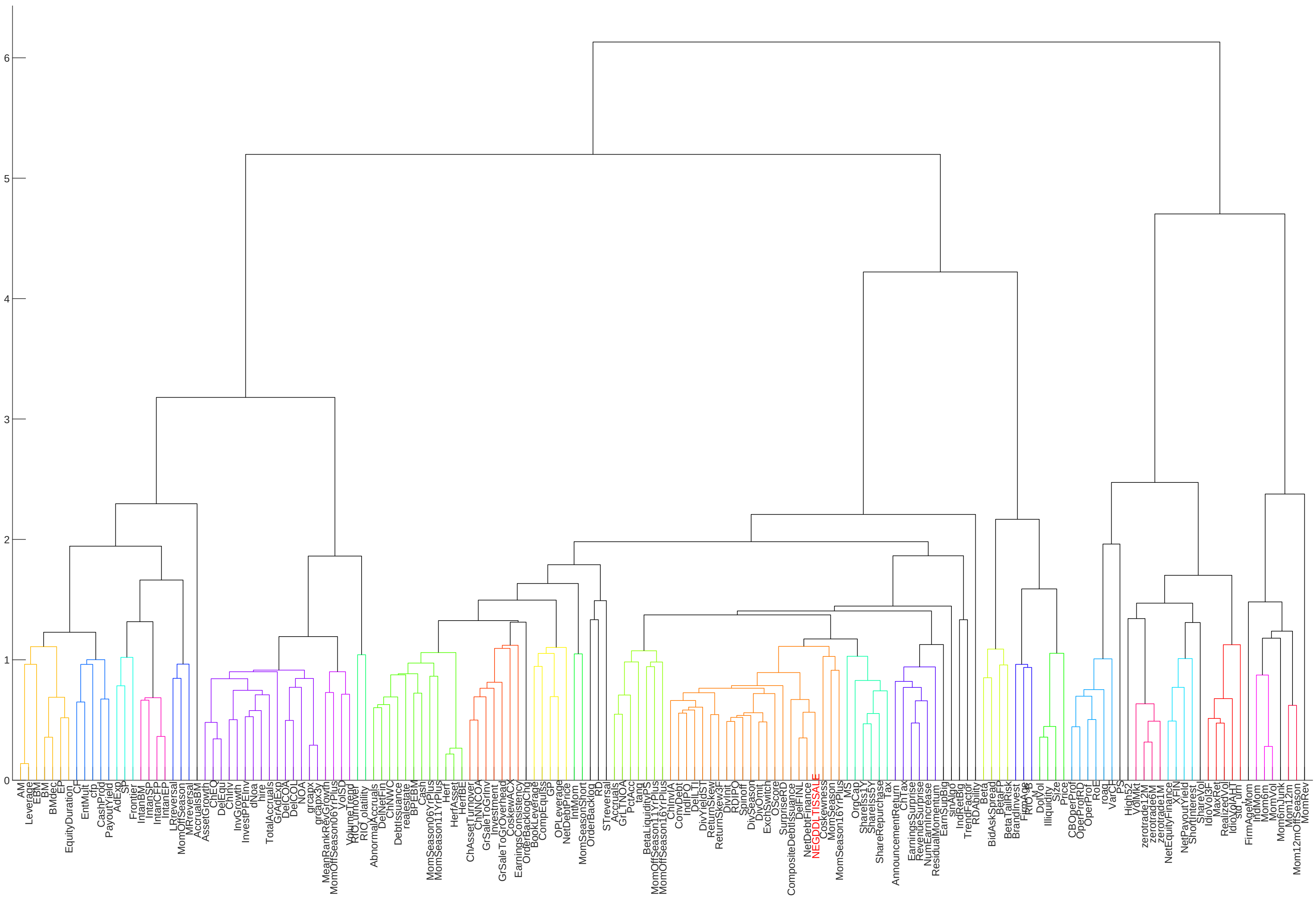


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

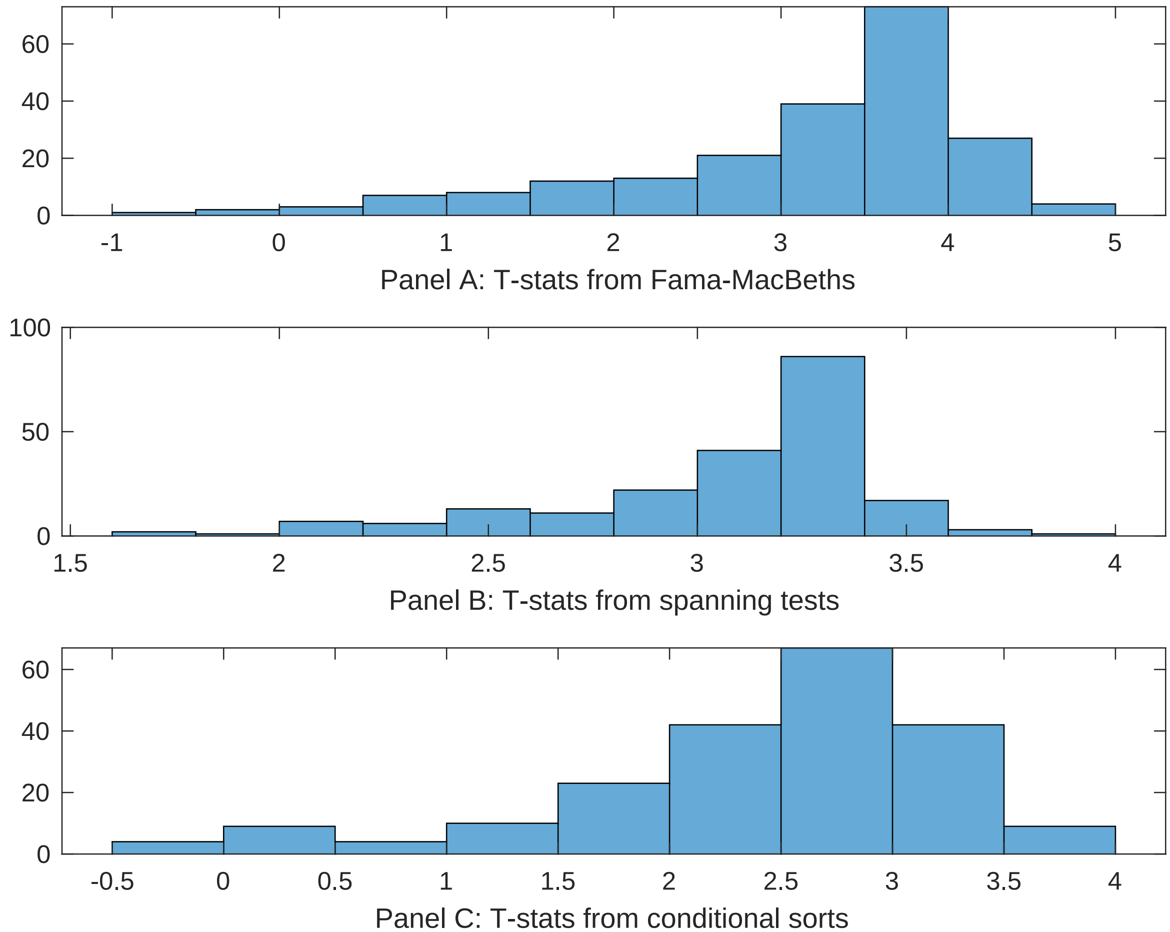


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD} RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDD} RDD_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.13	0.15
	[5.42]	[5.46]	[5.70]	[5.32]	[5.89]	[5.42]	[5.68]
RDD	0.95	0.89	0.68	0.29	-0.13	0.19	-0.54
	[0.14]	[0.01]	[1.08]	[3.55]	[-0.19]	[2.95]	[-0.66]
Anomaly 1	0.21						0.96
	[9.63]						[1.45]
Anomaly 2		0.18					-0.53
		[9.44]					[-1.14]
Anomaly 3			0.20				0.98
			[6.75]				[1.68]
Anomaly 4				0.40			0.91
				[6.87]			[1.63]
Anomaly 5					0.11		0.76
					[8.82]		[3.66]
Anomaly 6						0.94	0.81
						[6.22]	[0.58]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the RDD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDD} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15	0.15	0.15	0.17	0.17	0.19	0.15
	[2.28]	[2.22]	[2.23]	[2.54]	[2.45]	[2.73]	[2.24]
Anomaly 1	20.41						11.97
	[5.60]						[2.40]
Anomaly 2		17.17					7.28
		[4.51]					[1.45]
Anomaly 3			14.69				11.23
			[4.36]				[3.12]
Anomaly 4				6.28			4.45
				[2.37]			[1.63]
Anomaly 5					5.44		-1.81
					[1.38]		[-0.44]
Anomaly 6						6.94	3.61
						[1.89]	[0.94]
mkt	-3.56	-3.21	-1.59	-3.72	-3.61	-3.42	-1.83
	[-2.33]	[-2.07]	[-0.98]	[-2.38]	[-2.30]	[-2.18]	[-1.13]
smb	1.21	1.25	7.73	3.70	2.62	3.48	5.67
	[0.52]	[0.53]	[3.00]	[1.55]	[1.10]	[1.46]	[2.08]
hml	-18.60	-18.29	-17.80	-19.62	-19.67	-20.76	-18.02
	[-6.29]	[-6.12]	[-5.92]	[-6.50]	[-6.49]	[-6.69]	[-5.90]
rmw	-0.80	-0.10	-7.33	2.36	1.47	1.76	-6.43
	[-0.26]	[-0.03]	[-2.01]	[0.76]	[0.48]	[0.57]	[-1.76]
cma	30.13	29.67	25.85	29.83	28.82	29.23	17.62
	[6.57]	[6.31]	[5.13]	[5.80]	[4.35]	[5.19]	[2.56]
umd	2.36	2.19	3.58	3.04	3.80	3.23	1.45
	[1.53]	[1.40]	[2.33]	[1.92]	[2.43]	[2.05]	[0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	17	16	15	14	14	20

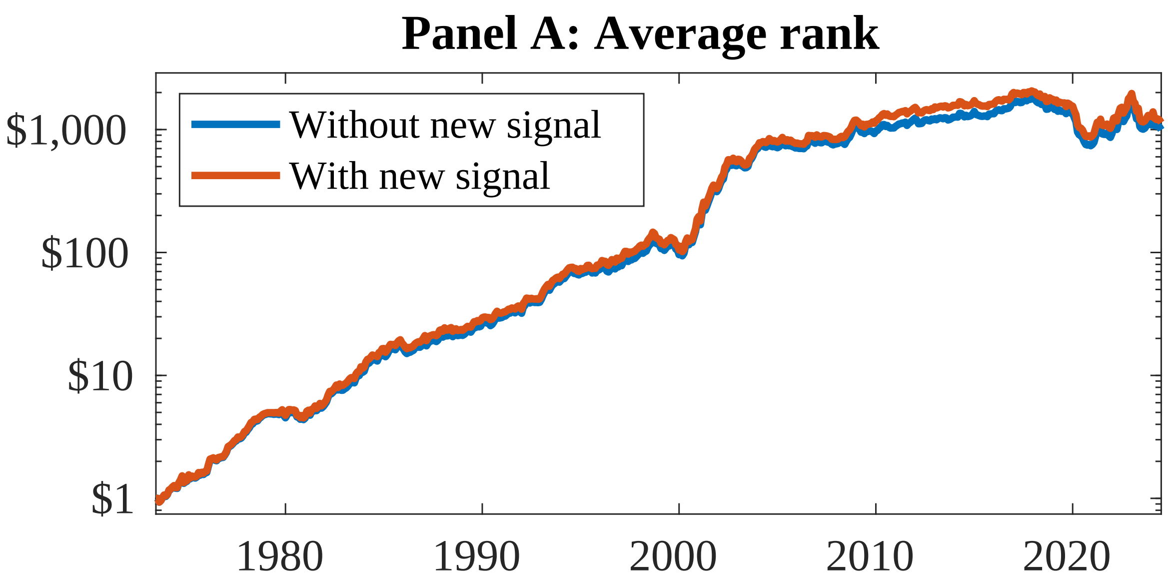


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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